NO.:GJW20261015049

# Material Safety Data Sheet

# 材料安全数据表

样　品　名　称:　锂离子电池

Name of Sample：　Li-ion Battery

委　托　单　位：　深圳凯万科技有限公司

Commissioner：　Shenzhen Cowon Technology Limited

For and on behalf of
Shenzhen Cowon Technology Limited
深圳市凯萬科技有限公司
Authorized Signature(s)

# 材料安全数据表

# Material Safety Data Sheet

| 1. 化学品及企业标识<br>Chemical product and company identification | |
|---|---|
| 样品名称<br>Name of Sample | 锂离子电池<br>Li-ion Battery |
| 样品型号<br>Type/Model | INR18500 26N |
| 产品参数<br>Ratings | 2600mAh 3.7V 9.62Wh |
| 委托单位<br>Commissioned by | 深圳凯万科技有限公司<br>SHENZHEN COWON TECHNOLOGY LIMITED |
| 委托单位地址<br>Commissioner address | 深圳市宝安区福永街道流塘路好运来大厦B栋F7016<br>F7, Haoyunlai Building B, Liutang, Xixiang, Baoan District, Shenzhen, China |
| 生产单位<br>Manufacturer | 深圳凯万科技有限公司<br>SHENZHEN COWON TECHNOLOGY LIMITED |
| 生产单位地址<br>Manufacturer address | 深圳市宝安区福永街道流塘路好运来大厦B栋F7016<br>F7, Haoyunlai Building B, Liutang, Xixiang, Baoan District, Shenzhen, China |
| 鉴定依据<br>Inspection according to | EEC Directive 93/112/EC<br>联合国《关于危险品货物运输的建议书》<br>UN “Recommendations on the TRANSPORT OF DANGEROUS GOODS” |
| 应急电话<br>Emergency telephone | 0753-5400318 |
| | 送样日期：2026-10-14　　签发日期：2026-10-15<br>Receiving date: 2026-10-14　Date of issue :2026-10-15 |

Approved By ：　　Reviewed By:　　Tested By :

批注：　覃爱娟　　审核：　胡记远　　主检：　王丽金

## 2. 成分组成信息
## Composition information

| 材料及组分<br>Chemical Name | 化学式<br>Chemical Formula | CAS 号<br>CAS NO. | 重量含量<br>Wt% |
|---|---|---|---|
| 镍钴锰酸锂 Lithium Ni-Co-Mn Oxid | LiNiCoMnO2 | 182442-95-1 | 39.6 |
| 丁苯橡胶<br>SBR | (C8H8.C4H6)X | 9003-55-8 | 1.78 |
| (PE)聚乙烯<br>Polyethylene | (CH2H4)n | 9002-88-4 | 0.06 |
| 聚偏二氟乙烯<br>PVDF | (CH2CF2)n | 24937-79-9 | 1.15 |
| 石墨<br>Graphite | C | 7782-42-5 | 23.2 |
| 六氟磷酸锂<br>Lithium Hexafluorophoshate | LiPF6 | 21324-40-3 | 15.35 |
| 碳酸乙烯酯<br>Ethylene carbonate | C3H4O3 | 96-49-1 | 2.72 |
| 铜箔<br>Copper Foil | Cu | 7440-50-8 | 9.8 |
| 铝箔<br>Aluminum Foil | Al | 7429-90-5 | 5.56 |
| (PP)聚丙烯<br>Polypropylene | (CH3H6)n | 9003-07-0 | 0.78 |
| 铅<br>Lead | Pb | 7439-92-1 | 无<br>Not Detected |
| 镉<br>Cadmium | Cd | 7440-43-9 | 无<br>Not Detected |
| 汞<br>Mercury | Hg | 7439-97-6 | 无<br>Not Detected |

| 3.危险性概述<br>Hazards Identification | |
|---|---|
| 爆炸危险性<br>Explosive Risk | 该物品不属于爆炸危险品<br>This article does not belong to the explosion dangerous goods . |
| 易燃危险性<br>Flammable Risk | 该物品不属于易燃危险品<br>This article does not belong to the flammable material. |
| 氧化危险性<br>Oxidation Risk | 该物品不属于氧化危险品<br>This article does not belong to the oxidation of dangerous goods. |
| 毒害危险性<br>Toxic Risk | 该物品不属于毒害危险品<br>This article does not belong to the toxic dangerous goods . |
| 放射危险性<br>Radioactive Risk | 该物品不属于放射性危险品<br>This article does not belong to the radiation of dangerous goods . |
| 腐蚀危险性<br>Mordant Risk | 该物品不属于腐蚀危险品<br>This article does not belong to the corrosion of dangerous goods. |
| 其他危险性<br>Other Risk | 该物品为锂离子电池，瓦时率 9.62Wh，属于锂离子电池（包括锂聚合物电池）<br>This article is Li-ion Battery, Watt hour rate 9.62Wh, Which belong to the Lithium ion batteries (including lithium polymer batteries) |

## 4. 急救措施
## First aid measures

眼睛：万一接触，立即用大量的清水冲洗至少 15 分钟，泛起上下眼脸。直到化学的残留物消失为止，迅速就医。

Eye：Flush eyes with plenty of water for at least 15 minutes,occasionally lifting the upper and lower eyelids. Get medical aid.

皮肤：万一接触，用大量水冲洗至少 15 分钟，同时除去污染的衣物和鞋子，迅速就医。

Skin：Remove contaminated clothes and rinse skin with plenty of water or shower for 15 minutes.Get medical aid .

吸入：立即从暴露处移至空气清新处，如果呼吸困难给及输氧，立即就医。

Inhalation: Remove from exposure and move to fresh air immediately.Use oxygen if available .

食入：饮用两杯牛奶或水。如果当事人仍然清醒可以采取催吐的方法，并且立即就医。

Ingestion: Give at least 2 glasses of milk or water .Induce vomiting unless patient is unconscious. Call a physician .

## 5. 消防措施
## Fire-fighting measures

燃点：不适用
Flash Point: N/A
自燃温度：不适用
Anto-Ignition Temperature: N/A
灭火介质：大量水（降温），二氧化碳
Extinguishing Media：Water .$CO_2$
特殊灭火程序：自给式呼吸器
Special Fire-Fighting Procedures : Self-contained breathing apparatus .
异常火灾或爆炸：当电芯暴露于过热的环境中时，安全阀可能会打开。
Unusual Fire and Explosion Hazards：
Cell may vent when subjected to excessive heat-exposing battery contents .
燃烧产生的危险物品：一氧化碳，二氧化碳，锂氧化物烟气
Hazardous Combustion Products :
Carbon monoxide. Carbon dioxide, lithium oxide fumes .

## 6. 泄露应急处理
## Accidental release measures

**为防止电池材料泄露或释放采取的措施**
如果电池内部材料泄露，实验人员应立刻撤离试验区直到烟气消散。将通风设备打开吹散危险性气体。用抹布擦净试验区，清除溢出的液体，将泄露电池放进塑料袋中，然后放进钢制容器。避免皮肤和眼睛接触或吸入有害气体。
Steps to be taken in case Material is Released or Spilled
If the battery material is released , remove the personnel form area until fumes dissipate.Provide maximum ventilation to clear out hazardous gases.Wipe it up with a cloth, and dispose of it in a plastic bag and put into a steel can .The preferred response is to leave the area and allow the battery to cool and vapors to dissipate. Provide maximum ventilation avoid skin and eye contact or inhalation of vapors. Remove spilled liquid with absorbent and incinerate .
**废弃物处置方法**
建议将电池完全放电，消耗电池内部的锂金属，并且深埋于土壤中。
**Waste Disposal Method**
It is recommended to discharge the battery to the end ,to use up the metal lithium inside the battery, and to bury the discharged battery in soil .

## 7. 操作处置和储存
## Handing and Storage

禁止打开、毁坏活焚烧电池，因为电池有可能在这些处理过程中发生爆炸、破裂或者泄露等事故。禁止将电池短路、过充、强制放电或扔入火中。禁止挤压刺穿电池或将电池浸入溶液中。

The battery should not be opened, destroyed or incinerate,since they may leak or rupture and release to the environment the ingredients that they contain in the hermetically sealed container.Do not short circuit terminals , or over charge the battery, forced over-discharge, throw to fire. Do not crush or puncture the battery ,or immerse in liquids.

**操作处置和储存中的防范措施**

禁止物理或电滥用，禁止高温储存，最好将电池储存在阴凉、干燥、通风及温度变化较小的环境中，禁止将电池接触加热设备或电池直接暴露与阳光中。

**Precautions to be taken in handing and storing .**

Avoid mechanical or electrical abuse. Storage preferably in cool ,dry and ventilated area, which is subject to little temperature change. Storage at high temperatures should be avoided. Do not place the battery near heating equipment , nor expose to direct sunlight for long periods .

其他要注意的防范措施

拆解、挤压、直接放入火中或高温条件下，电池可能发生爆炸和燃烧。禁止短接或电池正负极错误的安装在设备中。

Other Precautions

The battery may explode or cause bums, if disassembled , crushed or exposed to fire or high temperatures. Do not short or install with incorrect polarity.

## 8. 接触控制/个人防护
## Exposure controls/personal protection

**呼吸防护：**

当电池排气阀打开时，应尽量使通风设备开至最大，避免将打开排气阀的电芯局限在某一狭窄空间内。正常操作条件下。呼吸保护是不必要的。

Respiratory Protection

In case of battery venting,provide as much ventilation as possible.Avoid confined areas with venting cell cores. Respiratory protection is not necessary under conditions of normal use.

**通风条件**

正常使用条件下不必考虑

Ventilation

No necessary under conditions of normal use.

**防护手套**

正常使用条件下不必考虑

Protective Gloves

No necessary under conditions of normal use.
**其他防护服装或设备**
正常使用条件下不必考虑
Other Protective Clothing or Equipment
No necessary under conditions of normal use.
电池开阀试验时应做好个人防护
呼吸防护、防护手套、防护服装和有护边的安全玻璃罩都是要准备的。
Personal Protection is recommended for venting battery
Respiratory Protection, Protective Gloves, Protective Clothing and safety glass with side shields .

## 9. 物理和化学特性
## Physical and chemical properties

**外形：**圆柱形
Appearance: Cylindrical
**认证编号：**RZUN2021-1427
Ref,NO.:RZUN2021-1427
**气味：**泄露时，有醚的气味
O dour: If leaking, smells of medical ether
**酸碱度：**不适用
PH: Not applicable as supplied
燃点：除单个电芯暴露试验外其他不适用。
Flash Point: Not applicable unless individual components exposed .
**可燃性：**除单个电芯暴露试验外其他不适用。
Flammability:Not applicable unless individual components exposed .
**相对密度：**除单个电芯暴露试验外其他不适用。Relative density: Not applicable unless individual components exposed .
**溶解性（水溶性）：**除单个电芯暴露试验外其他不适用。
Solubility(water):Not applicable unless individual components exposed .
**溶解性（其他）：**除单个电芯暴露试验外其他不适用。
Solubility(other):Not applicable unless individual components exposed .

## 10. 稳定性和反应活性
## Stability and reactivity

**稳定性：**产品在第 7 节所述的条件下稳定。
Stability: Product is stable under conditions described in Section 7.
**应避免的条件：**加热 70℃以上或焚烧、变形、毁坏、粉碎、拆卸、过充电、短路、长时间暴露在

潮湿的条件下.
Conditions to Avoid: Heat above 70℃ or incinerate .Deform,Mutilate,Crush,Disassemble, Overcharge,Short circuit,Expose over a long period to humid conditions.
**应避免的材料：**氧化剂、碱、水
Materials to avoid: Oxidising agents, alkalis, water
**危险分解物：**不适用
Hazardous Polymerization: N/A
如果发生泄露，避免与强氧化剂、无机酸、强碱、卤代烃接触
If leaked ,forbidden to contact with strong oxidizers,mineral acids,strong alkalies, halogenated hydrocarbons.

## 11. 毒理学资料
## Toxicological information

**标志及症状：**无，除非电池破裂。
Signs & symptoms: None ,unless battery ruptures.
内部物质暴露的情况下，蒸汽烟雾可能对眼睛和皮肤的刺激性。
In the event of exposure to internal contents ,vapour fumes may be very irritating to the eyes and skin .
**吸入**:对肺有刺激性
Inhalation: Lung irritant
**皮肤接触：**对皮肤有刺激性
Skin contact： Skin irritant
**眼睛接触：**对眼睛有刺激性
Eye contact: Eye irritant
**食入：**吞下中毒
Ingestion: Poisoning if swallowed
下列情况下会危险人员身体健康：如果与电池内部材料直接接触，皮肤可能会出现干燥
灼烧等轻微或严重的刺激，并且损坏靶器官的神经，肝脏和肾脏。
Medical conditions generally aggravated by exposure:In the event of exposure to internal contents ,moderate to server irritation, burning and dryness of the skin may occur, target organs nerves , liver and kidneys .

## 12. 生态学资料
## Ecological information

**对哺乳动物的影响：**目前未知
Mammalian effects: None know at present
**生态毒性：**目前未知

Eco-toxicity: None know at present
**生物体内累积：**慢慢的生物降解
Bio-accumulation potential: Slowly Bio-accumulation
**环境危害：**目前没有已知的环境危害
Environmental fate: None know environmental hazards at present .

## 13. 废弃处置
## Disposal consideration

禁止焚烧，或使电池温度超过 70℃，这种滥用可导致泄漏或电池爆炸。应按照相应的地方性法规处理。
Do not incinerate or subject cells to temperature in excess of 70℃，such abuse can result in loss of seal leakage, and cell explosion .Dispose of in accordance with appropriate local regulations .

## 14. 运输信息
## Transport information

**运输标签：**锂电池标记，第九类危险品标签，仅限货机标签
Label for conveyance: Lithium battery Mark,Class 9 Hazard Label,Cargo Aircraft Oircraft Only Label
**UN 编号**: 3480
UN Number: 3480
**包装等级：**不适用
Packaging Group: N/A
**EMS 编号：** F-A-S-I
EMS NO.: F-A-S-I
**海洋污染物：** 无
Marine pollutant: NO
**正确装运名称：**锂离子电池（单独）运输（包括锂离子聚合物电池）
Proper Shipping name : Lithium ion battery (including lithium ion polymer battery)
**危害分类：**货物应遵守 IATA DGR(第 67 版)PI965 IB 规定或特殊规定 188 海运危险货物规则（Amdt.42-24)版，包括通过 UN38.3 测试。

Hazard classification: the goods shall comply with IATA DGR (67th Edition) PI965 Part IB or special provisions 188 maritime dangerous goods rules (amdt. 42-24), including UN38.3 test.

## 15. 法规信息
## Regulation information

法律信息
Law information
《危险物品规则》
《Dangerous Goods Regulations》
《对危险货物运输的有关规定的建议》
《Recommendations on the Transport of Dangerous Goods Model Regulations》
《国际海运危险货物规则》
《International Maritime Dangerous Goods》
《危险品安全运输技术指令》
《Technical Instructions for the Safe Transport of Dangerous Goods》
《危险货物分类和品名编号》
《Classification and code of Dangerous Goods》
《职业安全卫生法》
《Occupational Safety and Health Act》(OSHA)
《有毒物质控制法》
《Toxic Substance Control Act》(TSCA)
《消费产品安全法》
《Consumer Product Safety Act》(CPSA)
《联邦环境污染控制法》
《Federal Environmental Pollution Control Act》(FEPCA)
《石油污染法案》
《The Oil Pollution Act》(OPA)
《超级基金修正案和再授权法案Ⅲ(302/311/312/313)》
《Superfund Amendments and Re-authorization Act TitleⅢ(302/311/312/313)》(SARA)
《资源保护及恢复法案》
《Resource Conservation and Recovery Act》(RARA)
《安全饮用水法》
《Safety Drinking Water Act》(CWA)
《加州 65 提案》
《California Proposition 65》
《美国联邦法规》
《Code of Federal Regulations》(CFR)
根据所有联邦、州和地方法律
In accordance with all Federal,State and Local laws .

## 16. 其他信息
## Other information

本文件仅对由梅州市博富能科技有限公司提供的，并由梅州市博富能科技有限公司生产的单电芯电池（INR18500 26N）有效。该电池的成分信息由委托方提供并承诺其完整性和准确性。用户应仔细阅读此文件，并按照正确的方法使用电池。如因电池使用不当造成的损害或损失，梅州市博富能科技有限公司不承担任何责任。

This file is only effective to the batteries (INR18500 26N ) provided bu MEI ZHOU BO FU NENG TECHNOLOGY CO.,LTD which manufactured by MEI ZHOU BO FU NENG TECHNOLOGY CO.,LTD The commissioner provides the composition information of batteries, and promises it's integrity and accuracy. Users should read this file carefully and use the batteries in correct method .MEI ZHOU BO FU NENG TECHNOLOGY CO.,LTD doesn't assume responsibility for any damage or loss because of misuse of batteries .